



LOCUS BROWSER

Architecture & current features

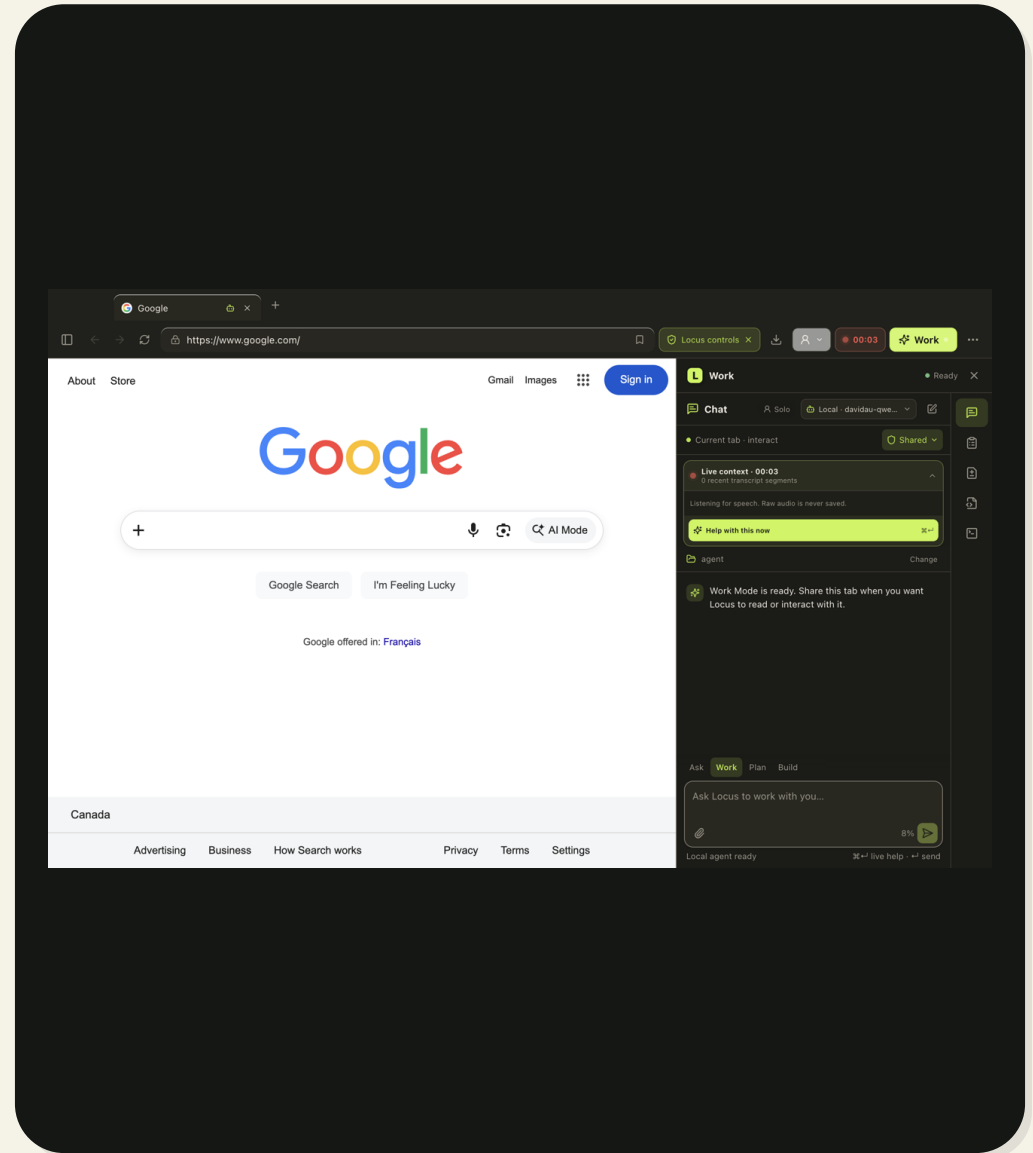
A browser-first desktop product with a focused solo agent, explicit tab sharing, visible live context, encrypted sync, and signed extension delivery.

macOS 14+

Apple Silicon

Electron + React

Canary



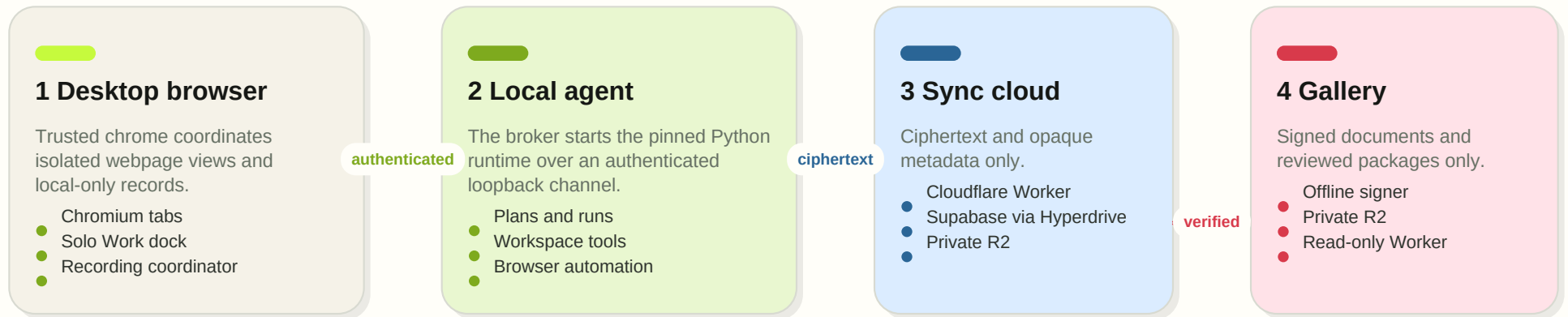
Implementation guide

Source-aligned as of 24 August 2026

Public canary architecture

The system at a glance

Four trust domains keep normal browsing, local agent work, encrypted sync, and extension delivery independently constrained.



Core design rule

Remote pages, models, cloud services, and extensions never receive ambient authority. Every privileged action is rechecked by the Electron main-process broker against the current sender, profile, conversation, tab grant, URL, and permission state.

Webpage is the primary canvas

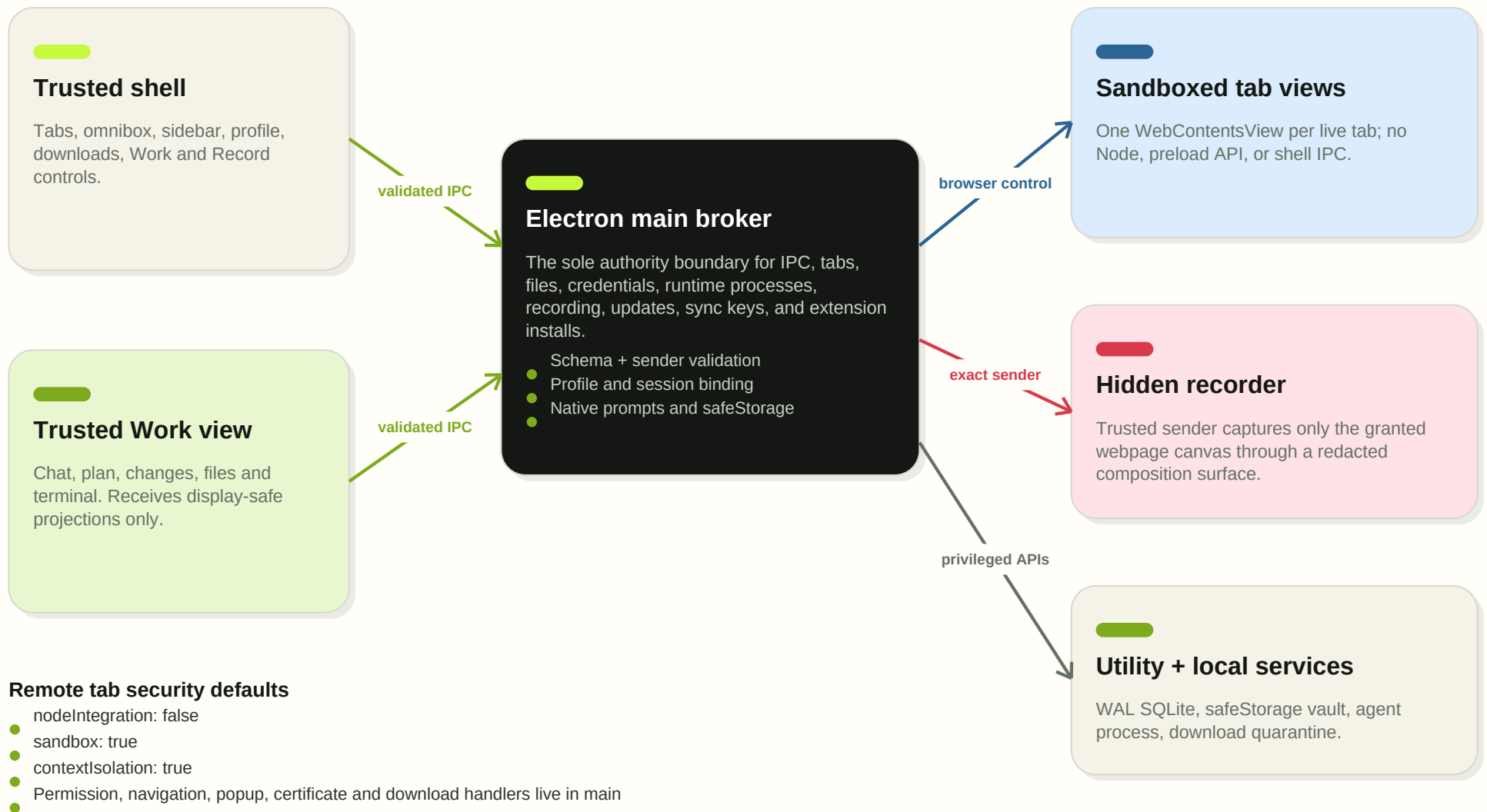
Work is optional

Cloud is optional

Private data stays local

Desktop processes and trust boundaries

A trusted shell never grants Node or privileged preload access to remote content.



A complete browser before Work opens

Work is closed at launch. The webpage remains primary and standard browser behavior stays conventional.

Tabs & windows

- Drag, group, restore
- Multi-window state
- Background sleeping

Profiles & privacy

- Partitioned sessions
- Ephemeral private windows
- Per-profile libraries

Navigation

- Omnibox + search choice
- Back, forward, reload
- Popup and certificate policy

Personal library

- History and bookmarks
- Downloads
- Browser sidebar

Page controls

- Find and zoom
- Print and PDF
- Media controls

Passwords

- OS-backed encryption
- Gesture-gated autofill
- Never exposed to agents

Reliability

- Crash/session restoration
- WAL browser database
- One-schema rollback

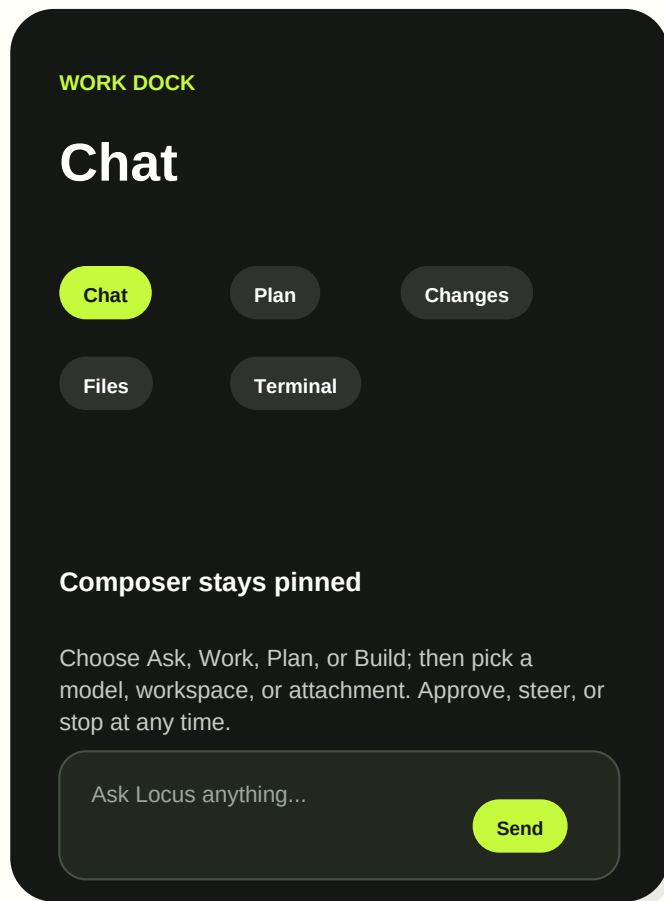
Accessibility

- Keyboard complete
- VoiceOver semantics
- Reduced Motion + 200% text

Performance target Warm tab switching under 150 ms at p95 on the Apple Silicon reference Mac.

Solo Work: the agent beside the page

The right dock is a focused agent workspace, not a second browser and not a team-orchestration console.



Conversation lifecycle

Create, list, resume, search, export, recover and keep active work running while the dock is hidden.

Workspace tools

Native folder choice, bounded files, live diffs, Git status, terminal events and permission prompts.

Agent control

Plan approval, action cards, evidence, progress, token context, steering, cancellation and runtime restart.

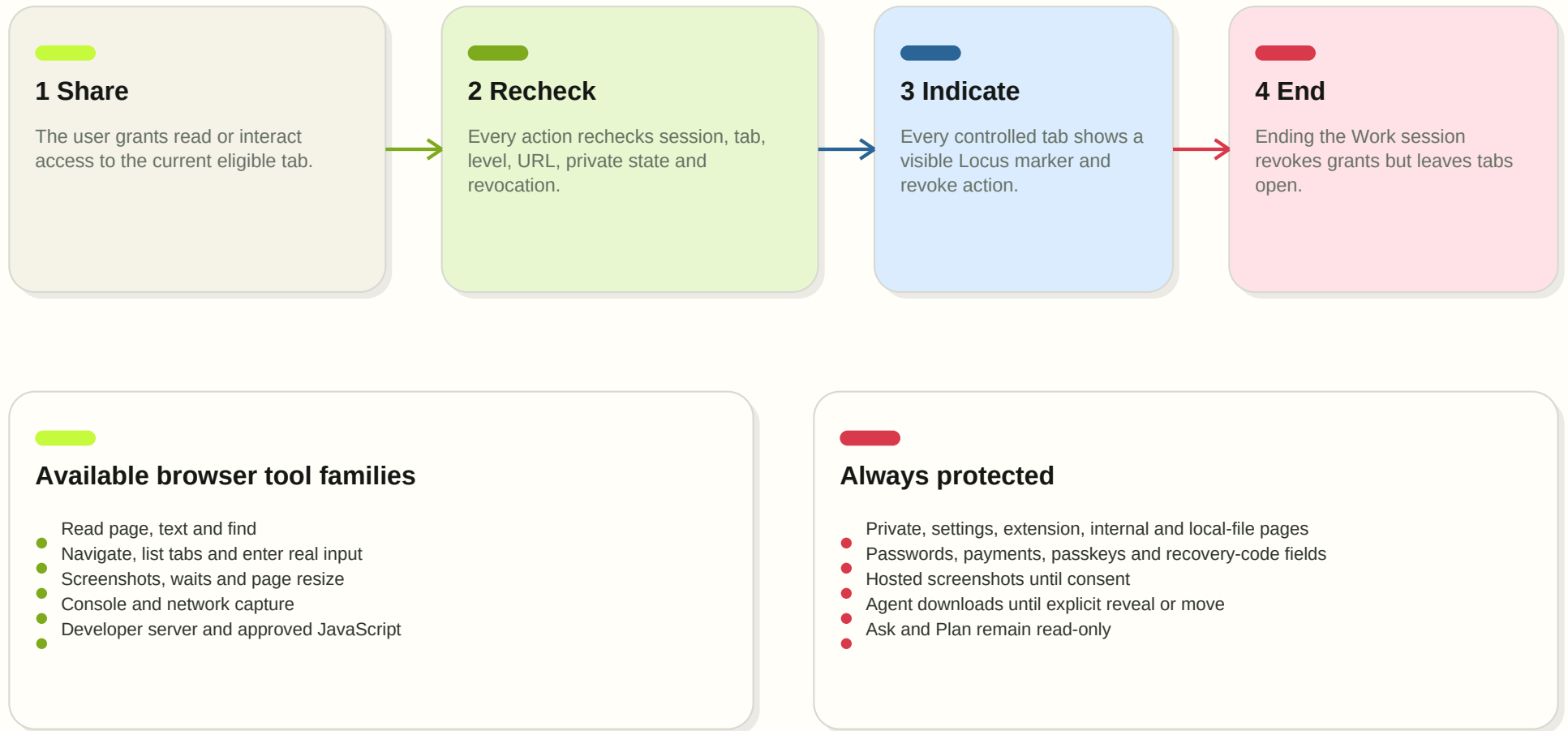
Model choice

- ChatGPT Plan
- ChatGPT API + Kimi
- Claude API
- vLLM / OpenAI-compatible
- Local Ollama models

Command+Option+L toggles Work without changing normal browser shortcuts.

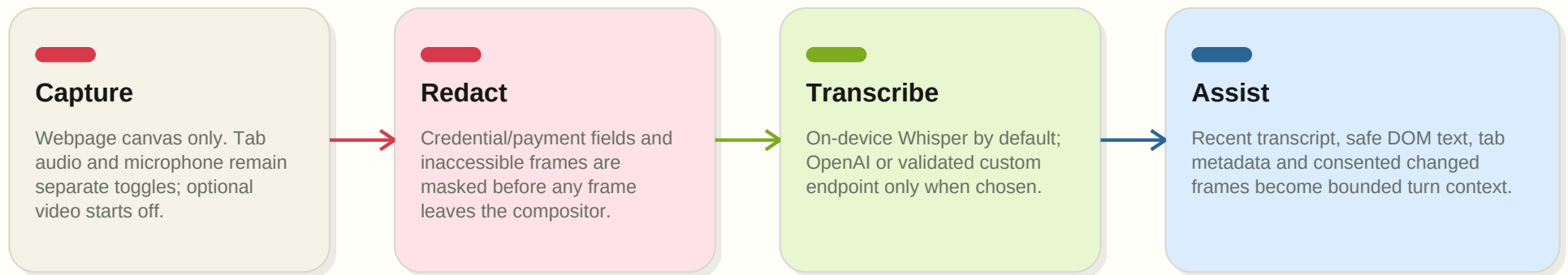
Browser control is granted, visible, and revocable

Tabs belong to the browser. A conversation receives only explicit access to a tab or ownership of a tab it created.



Live context: visible recording without ambient surveillance

Record follows only the active eligible shared tab and responds only when the user sends or presses Command+Enter.

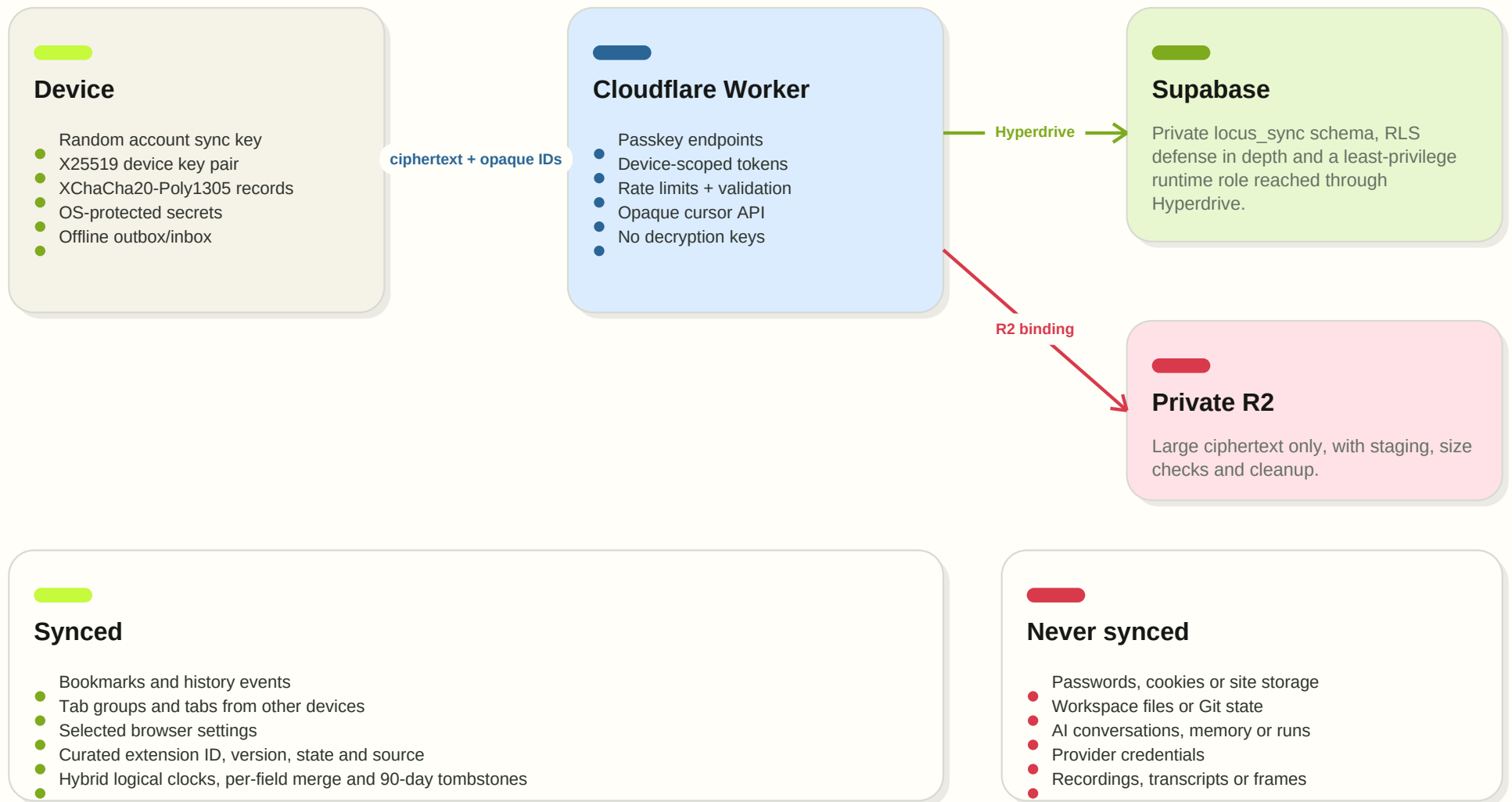


Privacy invariants

- The recording timer is always visible; there is no stealth mode.
- Private, internal, local-file, revoked, unshared, or stale-redaction tabs pause before another frame or audio chunk is accepted.
- Raw audio is never persisted. Transcripts are encrypted per record with a local key protected by macOS and never sync.
- Vision frames require hosted-screenshot consent. Optional redacted video is streamed locally and exported only after an explicit destination choice.

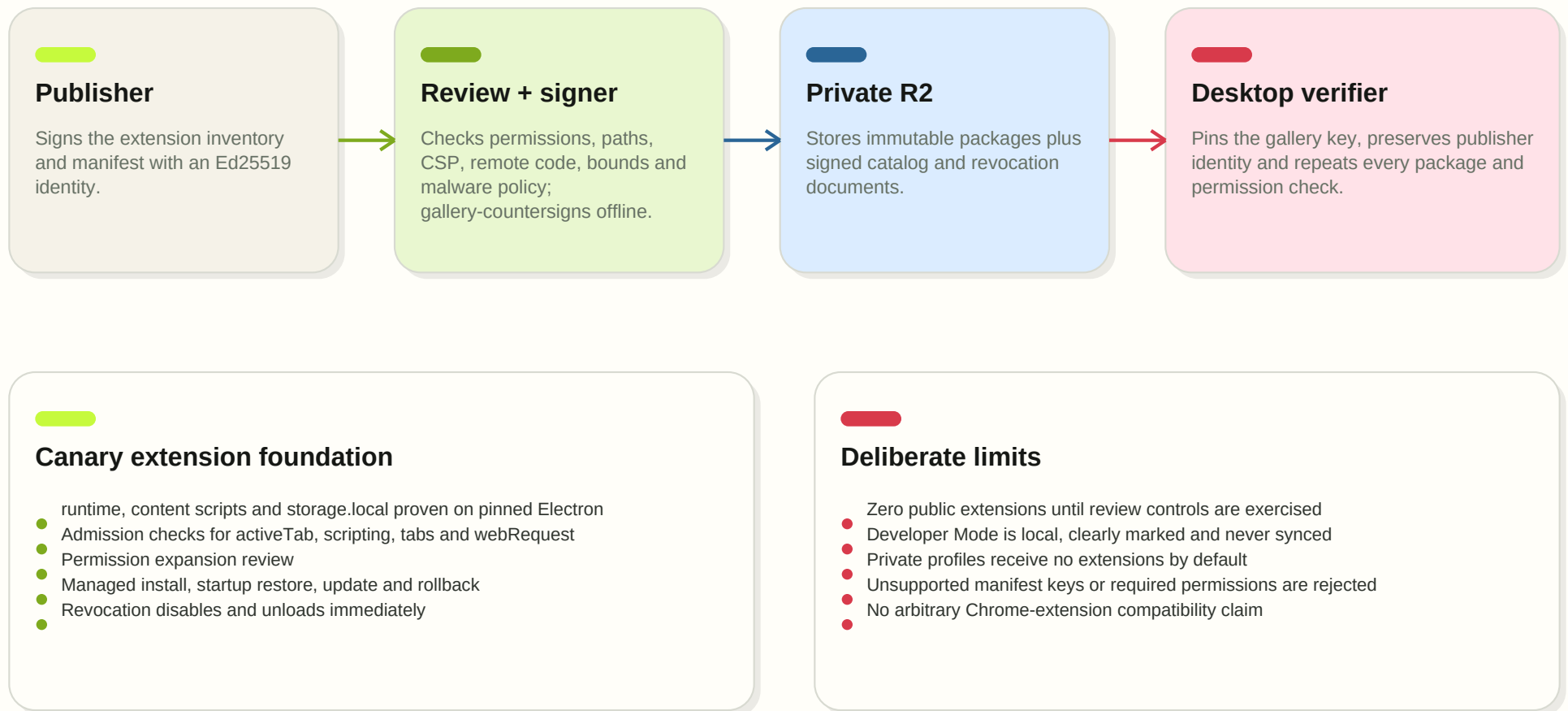
Encrypted sync: cloud storage cannot read content

Accounts are optional. Encryption and conflict handling happen on the device before Cloudflare receives data.



Extensions: curated compatibility, not Chrome Web Store claims

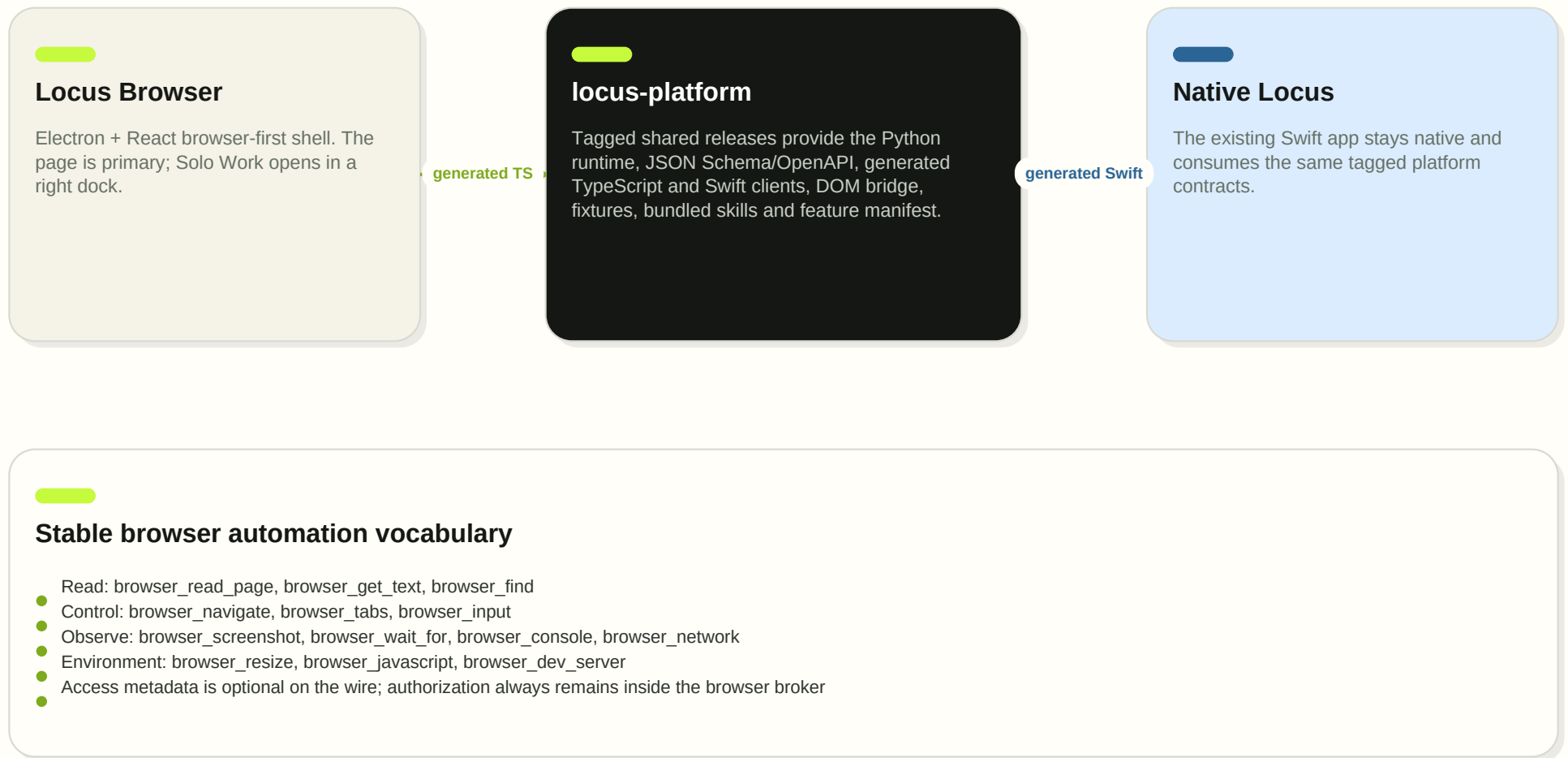
Locus owns a tested MV3 subset and a signed .locusx delivery contract.



Live gallery <https://extensions.locushost.co>

One shared platform, two native product surfaces

locus-platform keeps agent behavior and wire contracts engine-neutral while each app remains native to its product shape.



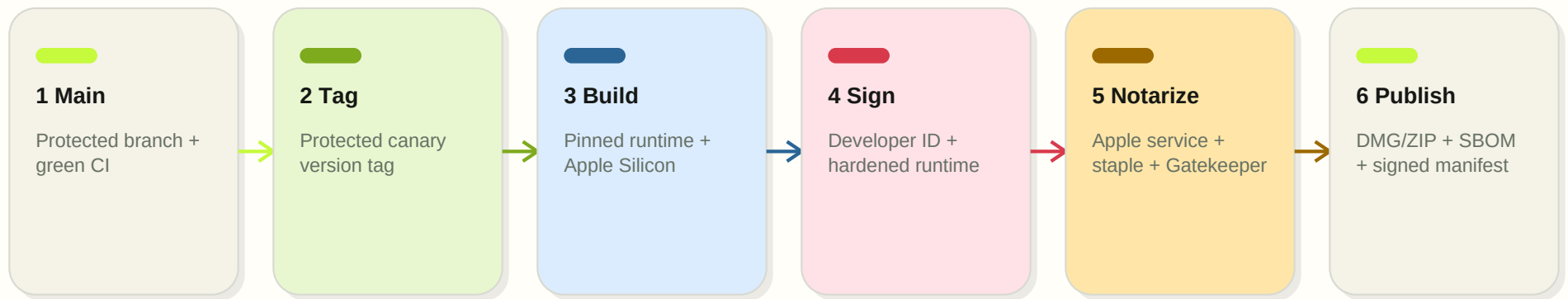
Where each kind of data lives

The architecture separates browser records, secrets, agent state, recordings, synced ciphertext, and public artifacts.

Data	Primary store	Boundary	Important rule
Browser records	WAL SQLite + Chromium profile partition	Local	Bookmarks/history/settings may sync encrypted
Passwords + provider keys	safeStorage / macOS Keychain	Local	Gesture-gated; never agent or sync data
Agent conversations + workspace	Authenticated local runtime	Local	Not browser sync data
Transcripts + video metadata	Encrypted record store + chosen file	Local	Raw audio never persists
Sync records	Supabase + private R2	Ciphertext	Service has no account sync key
Extension publication	Private R2 behind read-only Worker	Public verified	Signed catalog, revocations and packages
Release artifacts	GitHub protected canary workflow	Public signed	Notarized DMG/ZIP, SBOM and signed manifest

Canary delivery pipeline and current readiness

The source, cloud services, and trust roots are in place. The first public binary is held behind the protected release workflow.



Completed now

- Production sync Worker, Supabase/Hyperdrive and private R2
- Production extension Worker and private R2
- Separate gallery and release signing identities
- Pinned platform/runtime contracts and complete source gate
- Public architecture, privacy and release documentation

Remaining external launch evidence

- Enter the Developer ID certificate export password in the protected environment
- Run the notarized tag workflow and clean-Mac acceptance
- Complete independent security, sync-crypto and accessibility reviews
- Pass 24-hour internal cohort, soak, backup/restore and update/rollback rehearsals

SOURCE-COMPLETE CANARY CANDIDATE

No public binary is claimed until every external gate above is evidenced.